

Lamp Controller: User Guide

Welcome to the AI Hand Gesture Lamp Controller! This system allows you to completely control your physical environment using intuitive hand gestures picked up by your camera.

Follow this guide to master the controls.

1. Turning the Lamp ON



TURN ON
GESTURE

Gesture: Open Hand **Instruction:** To turn the lamp on, hold your hand up with all 5 fingers fully extended and separated. Hold the gesture steady for half a second to confirm.

2. Turning the Lamp OFF



Gesture: Closed Fist **Instruction:** To turn the lamp off, curl all of your fingers inwards to form a solid fist. Hold the gesture steady for half a second to confirm.

3. Adjusting Brightness



Gesture: Pinch **Instruction:** Ensure the lamp is ON. Pinch your thumb and index finger together, keeping your other three fingers completely closed. -
Increase Brightness: Spread your thumb and index finger apart. -
Decrease Brightness: Bring your thumb and index finger closer together.
The system uses a moving average filter, so the brightness will transition smoothly!

4. Panning the Lamp (Servo Control)



Gesture: Point **Instruction:** Ensure the lamp is ON. Extend your index finger while keeping your thumb, middle, ring, and pinky fingers completely closed. - **Move Left/Right:** With your index finger pointing up, move your entire hand left or right across the camera's view. The servo motor will smoothly pan the lamp to follow your hand's position horizontally (from 0° to 180°).

[!TIP] If the camera gets stuck or the system freezes, simply run `make stop` in your terminal to safely reset the background services before running `make run-lamp` again.